Strict Error Control in Physics-Informed Neural Networks via the Hypercircle Method

Xuefeng Liu

Tokyo Woman's Christian University

In collaboration with:

PingTing Hsieh & ShengYao Huang

(National Cheng Kung University, Taiwan)

March 2026

Outline

1. Motivation
2. Problem Setup
3. Hypercircle Theorem
4. Function Spaces: PINN vs. FEM
5. Admissible Neural Architectures
6. Where the Hypercircle Bound Is Not Strict
7. Singularity Enrichment
8. Numerical Experiments
9. Summary

Motivation: The Missing Guarantee in PINNs

What PINNs promise

- Mesh-free, flexible PDE solvers
- Works for forward & inverse problems
- No geometry-dependent meshing

Two critical bottlenecks

1. **No certified error bound**: residual is minimized *statistically*; no rigorous guarantee on $\|u - v_\theta\|$
2. **Costly 2nd-order derivatives**: computing Δv via autograd is expensive & exacerbates spectral bias

Our Goal

Derive a **computable, strict** *a posteriori* upper bound on the PDE approximation error — without knowing the exact solution u.

Key Idea

Revive the classical **Prager–Synge Hypercircle Theorem** (1947) and realise it via neural architectures that enforce admissibility *by construction*.

Problem Setup: Poisson Equation on Two Domains

PDE: Find u such that

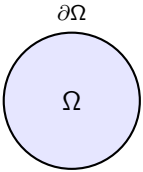
$$-\Delta u = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega.$$

Domain 1 — Unit Disk

$\Omega = \{(x, y) : x^2 + y^2 < 1\}$, $f \equiv 1$

$$u_{\text{exact}} = \frac{1-r^2}{4}, \quad \nabla u = \left(-\frac{x}{2}, -\frac{y}{2}\right)$$

Smooth solution, classical benchmark.

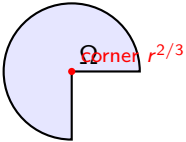


Domain 2 — L-shaped Sector

$\Omega = \{(r, \theta) : 0 < r < 1,\ 0 < \theta < \frac{3\pi}{2}\}$

$$u_{\text{exact}} = (r-1)^2 r^{2/3} \sin\left(\frac{2\theta}{3}\right)$$

Re-entrant corner: gradient singular at $r = 0$.



The Prager–Synge Hypercircle Theorem

Define two **admissible sets** in the Hilbert space $L^2(\Omega)^2$:

Kinematically Admissible $\mathcal{K}$

$v \in H_0^1(\Omega)$: satisfies Dirichlet BC

$$v = 0 \text{ on } \partial\Omega$$

Statically Admissible $\mathcal{S}$

$\mathbf{p} \in H(\mathrm{div}; \Omega)$: satisfies equilibrium

$$\mathrm{div}\,\mathbf{p} = -f \text{ in } \Omega$$

Orthogonality: For any $v \in \mathcal{K}$, $\mathbf{p} \in \mathcal{S}$, Green's identity gives

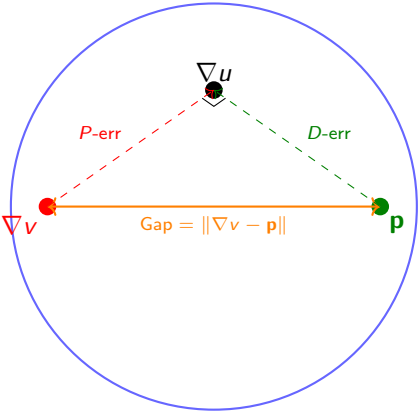
$$\langle \nabla v - \nabla u,\ \mathbf{p} - \nabla u \rangle = -\int_\Omega (v - u)\underbrace{\mathrm{div}(\mathbf{p} - \nabla u)}_{=0}\, d\Omega = 0.$$

Hypercircle Identity (Pythagorean Theorem in L^2)

$$\underbrace{\|\nabla v - \mathbf{p}\|^2}_{\textbf{Gap}^2\ \text{(computable!)}} = \underbrace{\|\nabla v - \nabla u\|^2}_{\text{Primal Error}^2} + \underbrace{\|\mathbf{p} - \nabla u\|^2}_{\text{Dual Error}^2}$$

$\Rightarrow \quad \|\nabla v - \nabla u\| \leq \|\nabla v - \mathbf{p}\|$ **(strict, computable upper bound)**

Geometric Interpretation: The Hypercircle



Key insight:

- ∇u lies on a *hypersphere* (circle in 2-D) whose diameter is the segment $[\nabla v, \mathbf{p}]$
- The right-angle at ∇u follows from orthogonality
- The diameter $\|\nabla v - \mathbf{p}\|$ is an **upper bound** for both primal and dual errors
- **No knowledge of u needed** to evaluate the bound!

Function Spaces: PINN vs. Classical FEM

Property	Classical FEM	PINN (this work)
Primal space (trial functions for u)		
Space	$V_h \subset H_0^1(\Omega)$, piecewise polynomial	$\mathcal{N}_v \cdot d(\mathbf{x})$, globally C^∞
BC satisfaction	Nodal interpolation / strong imposition	**Exact**: $v = d(\mathbf{x})\mathcal{N}_v$, $d\|_{\partial\Omega} = 0$
Approximation	Local (mesh-based), $O(h^k)$ convergence	Global (mesh-free), width/depth
Regularity	H^1 (potentially discontinuous gradient)	C^∞ (smooth activations like tanh
Dual space (flux field $\mathbf{p}$)		
Space	Raviart–Thomas $RT_k \subset H(\mathrm{div};\Omega)$	$\mathbf{p}_p + \mathrm{rot}(\mathcal{N}_\phi)$, globally C^∞
$\mathrm{div}\,\mathbf{p} = -f$	Exactly at element level (RT property)	**Exact**: $\mathrm{div}(\mathrm{rot}(\cdot)) \equiv 0$ by identity
Construction	Complex: local Neumann problems / patch recovery	Simple: one scalar NN + autogra
Regularity	Piecewise, $H(\mathrm{div})$ only	C^∞ flux field
Error bound		
Hypercircle bound	Requires expensive flux equilibration (Ainsworth–Oden)	**Direct**: minimise $\|\nabla v - \mathbf{p}\|^2$

Primal Network: Kinematic Admissibility by Design

Goal: ensure $v \in H_0^1(\Omega)$, i.e. $v = 0$ exactly on $\partial\Omega$.

Hard boundary condition via distance mask:

$$v(\mathbf{x}) = d(\mathbf{x}) \cdot \mathcal{N}_v(\mathbf{x}; \theta)$$

- $d(\mathbf{x})$ vanishes on $\partial\Omega$ analytically
- Regardless of weights θ: $v|_{\partial\Omega} = 0$
- $\mathcal{N}_v$: fully-connected, $2 \to 50 \to 50 \to 1$, tanh

Distance masks used:

$$d_{\text{disk}}(\mathbf{x}) = 1 - x^2 - y^2$$
$$d_{\text{sector}}(\mathbf{x}) = (1 - r)\,\theta\,(\tfrac{3\pi}{2} - \theta)$$

Admissibility Verification

$v = d \cdot \mathcal{N}_v \;\Rightarrow$

- BC: $v|_{\partial\Omega} = 0$ ✓
- Regularity: d bounded, $\mathcal{N}_v \in C^\infty$ $\Rightarrow v \in H^1(\Omega)$ ✓
- **Conclusion:** $v \in \mathcal{K}$

Note on Regularity

With tanh activations, $\mathcal{N}_v$ is C^∞. Multiplied by smooth d, v is smoother than any FEM piecewise polynomial.

Dual Network: Static Admissibility by Design

Goal: ensure $\operatorname{div} \mathbf{p} = -f$ *exactly in* Ω.

Decomposition:

$$\mathbf{p} = \underbrace{\mathbf{p}_p}_{\operatorname{div} \mathbf{p}_p = -f} + \underbrace{\operatorname{rot}(\mathcal{N}_\phi)}_{\operatorname{div}(\operatorname{rot}(\cdot)) \equiv 0}$$

where $\operatorname{rot}(\phi) = \big(\partial_y \phi,\ -\partial_x \phi\big)$.

Particular solutions:

- Disk ($f = 1$): $\mathbf{p}_p = (-\frac{x}{2}, -\frac{y}{2})$ (analytical)
- Sector (singular f): $\mathbf{p}_p = \sin(\frac{2\theta}{3})\big(2.5 r^{5/3} - 2.8 r^{2/3}\big)(\cos\theta, \sin\theta)$

Admissibility Verification

$$\operatorname{div} \mathbf{p} = \underbrace{\operatorname{div} \mathbf{p}_p}_{=-f} + \underbrace{\operatorname{div}(\operatorname{rot} \phi)}_{=0} = -f$$

Holds **algebraically**, for any θ.

Why curl works

Vector identity: $\operatorname{div}(\operatorname{rot}(\phi)) \equiv 0$ for any differentiable ϕ.

Same principle as *mimetic* / div-free finite element formulations.

Network: $\mathcal{N}_\phi$: $2 \to 50 \to 50 \to 1$, tanh; gradients via autograd $\Rightarrow \operatorname{rot}(\mathcal{N}_\phi)$.

The Hypotenuse Loss: Unified Joint Training

Separate training (baseline):

1. Primal: minimise PDE residual

$$\mathcal{L}_{\text{PDE}} = \|\Delta v + f\|^2 \quad \leftarrow \text{2nd-order derivatives!}$$

2. Dual (after primal converges):

$$\mathcal{L}_{\text{dual}} = \tfrac{1}{2}\|\mathbf{p}\|^2$$

Drawbacks: expensive Δv, spectral bias, loose coupling between v and $\mathbf{p}$.

Hypotenuse (joint) training (proposed):

$$\boxed{\mathcal{L}_{\text{gap}}(\theta, \eta) = \|\nabla v - \mathbf{p}\|^2}$$

- Only **1st-order derivatives** (∇v, $\operatorname{rot} \mathcal{N}_\phi$)
- Jointly optimises v and $\mathbf{p}$
- Loss value *is* the certified error bound
- Minimising $\mathcal{L}_{\text{gap}} \to 0$ forces both $v \to u$ and $\mathbf{p} \to \nabla u$ simultaneously

Strategy	Derivative order	Coupling	Certified bound?
Separate training	2nd	None	Yes (loose)
Hypotenuse (ours)	**1st**	**Joint**	**Yes (tight)**

Where the Hypercircle Bound Is *Not* Strict

The Pythagorean identity $G^2 = P^2 + D^2$ holds **only if** both admissibility conditions hold exactly. In practice, several sources of inexactness arise:

1. **Black-box or non-integrable source terms**
   When f has no closed-form antiderivative, an auxiliary network $\mathcal{N}_{\text{aux}}$ is pre-trained to minimise $\|\operatorname{div} \mathcal{N}_{\text{aux}} + f\|^2$. This gives only $\operatorname{div} \mathbf{p} \approx -f \Rightarrow$ static admissibility is violated; the bound $G^2 \geq P^2$ may fail.

2. **Monte-Carlo / numerical quadrature of L^2 norms**
   The squared norms G^2, P^2, D^2 are evaluated by *averaging over random collocation points*. Finite sample size introduces integration error $\Rightarrow$ identity discrepancy $\mathcal{E}_{\text{id}} = |G^2 - (P^2 + D^2)| > 0$ (observed: $\sim 10^{-9}$ in our tests).

3. **Floating-point regularisation near singularities**
   The numerical stabiliser $r \leftarrow \sqrt{x^2 + y^2 + \varepsilon}$ (with $\varepsilon = 10^{-12}$) used in the sector domain introduces a tiny perturbation to the distance mask and particular solution $\Rightarrow$ BC and divergence constraints hold only to $O(\sqrt{\varepsilon})$ near the corner.

4. **Non-smooth distance mask at corners**
   The sector mask $d = (1 - r)\,\theta\,(3\pi/2 - \theta)$ involves ∂_r at $r = 0$; the sector is not a C^1 domain $\Rightarrow$

Singularity Enrichment for the L-shaped Domain

Problem: re-entrant corner angle $\omega = 3\pi/2$.

Elliptic regularity (Grisvard 1985):

$$u \in H_0^1 \quad \text{but} \quad u \notin H^2, \qquad u \sim r^{\pi/\omega} = r^{2/3} \text{ as } r \to 0.$$

Standard tanh network cannot represent $r^{2/3}$ efficiently $\Rightarrow$ large errors near corner.

Solution — singular mask:

$$v(\mathbf{x}) = r^{2/3} \cdot d(\mathbf{x}) \cdot \mathcal{N}_v(\mathbf{x})$$

where $d = (1 - r)\,\theta\,(3\pi/2 - \theta)$.

The $r^{2/3}$ factor embeds the known asymptotic behaviour; $\mathcal{N}_v$ only needs to learn the *smooth remainder*.

Effect on admissibility

- $v|_{\partial\Omega} = 0$ still holds (distance mask d)
- $v \in H_0^1(\Omega)$ since $r^{2/3} \in H^1$ and $2/3 > 1/2$
- Gradient ∇v is now in L^2 by construction

Comparison of enrichment strategies

Strategy	P^2	G^2
No mask	2.92×10^{-3}	3.72×10^{-3}
With $r^{2/3}$	$\mathbf{4.03 \times 10^{-4}}$	$\mathbf{1.13 \times 10^{-3}}$

7.2$\times$ improvement in primal error

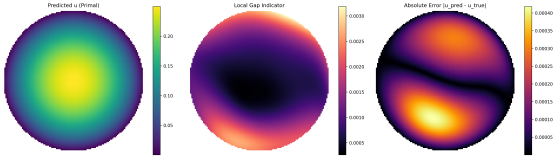
Results: Unit Disk ($-\Delta u = 1$)

Training: 8 000 samples, 5 000 epochs, Adam (lr $= 10^{-3}$), float64, CPU.

Method	**Time (s)**	G^2	P^2	D^2	$\mathcal{E}_{\text{id}}$
Hypotenuse (joint)	45.76	1.84×10^{-6}	5.57×10^{-7}	1.29×10^{-6}	4.25×10^{-9}
Separate	107.22	2.85×10^{-4}	6.83×10^{-6}	2.78×10^{-4}	1.42×10^{-7}

Key findings

- Hypotenuse: **155× tighter bound** (G^2 ratio: 2.85×10^{-4} vs 1.84×10^{-6})
- **2.3× faster** training (45 s vs 107 s)
- Identity discrepancy $\sim 10^{-9}$ (machine precision level)



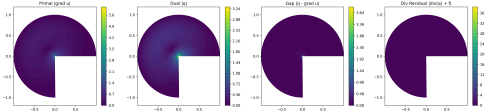
Results: L-shaped Sector (Singular Solution)

Training: 10 000 samples, 10 001 epochs, Adam (lr $= 10^{-3}$), float64, $2 \to 80 \to 80 \to 1$ networks, CPU.

Method	**Time (s)**	G^2	P^2	D^2	$\mathcal{E}_{\text{id}}$
Separate	501.22	1.01×10^{-1}	9.24×10^{-2}	8.81×10^{-3}	1.99×10^{-5}
Hypo, w/o mask	181.88	3.72×10^{-3}	2.92×10^{-3}	8.94×10^{-4}	8.99×10^{-5}
Hypo, w/ $r^{2/3}$	191.54	$\mathbf{1.13 \times 10^{-3}}$	$\mathbf{4.03 \times 10^{-4}}$	$\mathbf{7.28 \times 10^{-4}}$	7.24×10^{-7}

Key findings

- Joint training vs separate: **27×** **improvement** in G^2 ($10^{-1} \to 3.72 \times 10^{-3}$)
- Adding $r^{2/3}$ mask: additional **7.2×** reduction in P^2
- Hypotenuse bound remains **strict**: $G^2 = P^2 + D^2$ to $\mathcal{E}_{\text{id}} \sim 10^{-7}$
- Hypotenuse **2.6×** **faster** than separate training (191 s vs 501 s)



Summary of Contributions

1. Strictly Admissible Architectures

- Distance mask $\Rightarrow$ exact BC ($v \in \mathcal{K}$)
- Curl operator $\Rightarrow$ exact $\operatorname{div} \mathbf{p} = -f$ ($\mathbf{p} \in \mathcal{S}$)
- Hard constraints, no penalty terms

2. Hypotenuse Loss

- First-order only: no Δv needed
- Certified upper bound $G^2 \geq P^2, D^2$
- Joint training: faster & more accurate

3. Singularity Enrichment

- Asymptotic theory: $u \sim r^{\pi/\omega}$
- Explicit $r^{2/3}$ factor in trial space
- $7.2\times$ improvement in primal error

Future Directions

- 3-D: vector potentials for curl
- Adaptive sampling guided by local gap
- Extension to elasticity, Maxwell
- Automatic $\mathbf{p}_p$ construction for general f

Code: `https://github.com/ShengYaooo/Strict-Error-Control-in-Physics-Informed-Neural-Networks-via-the-Hypercircle-Method` (commit 727e802)

Thank You

Xuefeng Liu

TWCU

In collaboration with: PingTing Hsieh & ShengYao Huang (NCKU)

Hypercircle Identity:

$$\underbrace{\|\nabla v - \mathbf{p}\|^2}_{\text{computable gap}} = \|\nabla v - \nabla u\|^2 + \|\mathbf{p} - \nabla u\|^2$$